

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Test Method of Interfacial Tensiometer, Sigma 702ET (Biolin)

Equipment Number: EQ-03 and EQ-04
Serial Number: 72627 and 72630



1. PURPOSE

This standard operating procedure is intended to provide users with the guide on the operation and maintenance of Sigma 702ET from Biolin Scientific Interfacial Tensiometer.

2. SCOPE

This standard operating procedure is employed in the GLTS Analytical Laboratory primarily for performing interfacial tension determination of oil samples by ring method using Du Nouy tensiometer. As measurement probe a horizontally hanging platinum ring with known dimensions is used. Sigma 702ET model tensiometer uses a standard ring with an average ring radius of $R = 9,58 \text{ mm}$ and a wire radius $r = 0.185 \text{ mm}$. The platinum ring is immersed in the liquid and pulled out again. The value obtained from the measurement is the surface free energy of the interface between two immiscible, or poorly miscible liquids.

3. TERMS AND DEFINITIONS

None

4. RESPONSIBILITIES

- Use, care and maintenance of the Sigma 702ET Interfacial Tensiometer is directly assigned to the end user who is usually the Sr. laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. laboratory chemist.
- Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

Doc.: EOP [6]-04-E	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 1 of 5	

GCC Lab المختبر الخليجي	Gulf Laboratory Technical Services
	Test Method of Interfacial Tensiometer, Sigma 702ET (Biolin)

5. SAFETY PRECAUTIONS

To take the ring from the box removes the lid by slightly turning it. Take the opening between your fingers so that the ring can fall into the palm of your hand. Take the ring by the shaft with the other hand. The Du Nouy ring is a highly sensitive probe which gets useless when deformed. Don't touch the ring itself with your fingers. It can easily get deformed by even low forces.

6. PRINCIPLES OF OPERATION

- Measurement Preparations for measurement Before beginning a measurement, please go through the following checks and preparatory operations:
- Make sure that the transportation lock of the balance is unlocked.
- Check that the temperature and pressure of the room (15 – 30 C°) (10 – 15 psi)
- Turn on the Sigma instrument. The instrument should be switched on well before measurement to ensure stability. For the best results, switch it on 24 hours (or more) before measurement.
- Connect the data collection extension(s) you wish to use, such as a USB cable to PC and/or the optional printer.
- Calibrate the instrument if necessary (see section 5). When in doubt, do the calibrations.
- Check the condition of the measurement equipment you are going to use A correct, undeformed DuNoüy ring is important for measurements.
- Check the Water ST (Surface Tension) limit (ASTM D971) in the Adjust parameters menu. According to ASTM D971, the water surface tension has to be over 71 mN/m. If desired, the limit can be changed to accept a greater range or a smaller range of water surface tension values. When performing an interfacial tension measurement according to ASTM D971, the measurement cannot be continued if the measured water surface tension is below this set level. If you wish your measurements to be compliant with the ASTM D971 standard, keep the value at 71 mN/m.
- According to IEC 62961, the water surface tension must be between 70–73 mN/m. When performing an interfacial tension measurement according to IEC 62961, the measurement cannot be continued if the measured water surface tension is outside this range. When starting measurements, it is advisable to use gentle finger movements in pressing the buttons of the keypad, in order to avoid adding noise to the initial taring. The taring is performed by averaging balance readings over the period of a few seconds.

7. OPERATIONAL DETAILS

Performing an interfacial tension measurement between water/oil according to ASTM D971 standard

1. Check the instructions in (section 6) and make necessary preparations.

Doc.: EOP [6]-04-E	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 2 of 5	

GCC Lab المختبر الخليجي	Gulf Laboratory Technical Services
	Test Method of Interfacial Tensiometer, Sigma 702ET (Biolin)

2. Set oil density to correct value. You may also measure the density of the oil (see section 6).
3. Make sure that the correction method of your preference is selected. You can change the active ring correction method from the adjust parameters menu. To be strictly compliant with ASTM D971 standard, choose Zuidema-Waters.
4. Make sure the correct ring dimensions have been selected. Choose Adjust Parameters Ring dimensions and either choose the standard ring dimensions or use custom ring dimensions in case you have a custom ring.
5. Pour distilled water at a temperature of $25 \pm 1^{\circ}\text{C}$ into cleaned sample container to a depth of 10 to 15 mm. Place it into the middle of the tensiometer platform. Make sure no foam is present at the water surface. If foam is visible, discard the water. Foam indicates that the water is contaminated or that the vessel is not clean.
6. Mount the ring in the tensiometer hook. If the oil to be measured is denser than water, the measurement must be performed in Push Mode1, and the supplied weight therefore needs to be attached to the ring before mounting the ring.
7. Select Measure ASTM D971 from the menu selection and press the OK button. Sigma will now measure automatically the surface tension of the water. Wait until the measurement is completed. If the measurement was successful, the ring will automatically descend back into the water, except in the case of Push Mode measurement with dense oil, however, the ring will remain about 5 mm above the water.
8. When the ring has stopped moving, carefully layer the oil on the surface of the water, until an oil layer of at least 10 mm is reached. Do not move the sample container from the tensiometer platform, nor move the ring.
9. Press the OK button right after the oil pouring is finished. The tensiometer will then wait for 29 seconds (as the standard requires a wait between 29 and 31 seconds), until it starts the measurement of the interfacial tension between water and oil. Wait until the measurement is completed.
10. Interfacial tension value is shown on the display. As the ring is pulled through a surface or an interface, a curve is plotted, showing the tension value calculated from the balance reading at each instant. The highest value encountered will be displayed as the result of the measurement.

8. MEASURING DENSITY

8.1 Two ways to density measurement

NOTE: Density measurement is sensitive to vibrations. In order to ensure accurate measurements, the instrument should be placed on a vibration free table in an area where local air currents are at a minimum.

Doc.: EOP [6]-04-E	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 3 of 5	

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Test Method of Interfacial Tensiometer, Sigma 702ET (Biolin)

The density of transformer oil or other liquid can be measured using the glass probe supplied with the instrument. There are two alternative ways to start a density measurement, beginning from the main menu:

1. Select Measure surface tension and then Measure density or
2. Select Change oil density and then Measure oil density.

The difference between these is that with alternative 1, the result of the measurement becomes the default density value for surface tension measurement. With alternative 2, the result becomes the new oil density setting. However, in the case where the default density value for surface tension measurement is initially the same as the oil density setting, alternative 2 will copy the result value to both places. Whichever way is used, the result will be displayed on screen and sent to the PC receiver program and/or the optional printer if either of them is connected to the unit.

8.2 Density measurement steps the steps for performing the density measurement are as follows:

- Check the instructions in section 4 and make the necessary preparations.
- Place a sample container with sample liquid on the sample stage and carefully hang the density probe on the balance hook.
- Navigate the menu to density measurement, using either of the two ways described in section 6.1 then follow the instructions on screen:
 - Place the sample liquid on the sample stage. The depth of the liquid should be at least 17 mm, 2 cm to be sure. (The immersion depth of the density probe is about 14 mm.)
 - Hang the density probe on the balance hook.
 - Move the sample stage close to the probe to minimize waiting time during measurement.
- When the balance has stabilized, gently press OK gently to start the measurement.
- Wait for the completion of the measurement and write down the density reading. Density results may also be collected to data collector software on PC or an optional printer.

9. MEASURING SURFACE TENSION

The surface tension of a liquid can be measured using DuNoüy ring as follows:

- Check the instructions in section 6.1 and make all necessary preparations.
- Select Measure surface tension from the main menu. A density confirmation screen opens, asking you to confirm the density value or specify the correct density value. The

Doc.: EOP [6]-04-E	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 4 of 5	

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Test Method of Interfacial Tensiometer, Sigma 702ET (Biolin)

density can be given by density measurement or by modifying the density value manually. For density measurement, see section 6.

- When satisfied with the density, select Accept and continue from the density confirmation screen. The next screen will ask you about the number of measurement rounds, with your last choice as the default. Press the arrow buttons ↑ and ↓ to increase and decrease the round count and OK to continue.
- The display that follows instructs you to place the sample vessel on the stage, move the stage close to the probe. Move the stage with the stage movement buttons so that the surface of the liquid comes close to the DuNoüy ring but does not touch it. This way the waiting time at the beginning of the measurement will be minimized. When the balance has stabilized, gently press OK to begin the measurement.
- Wait until the measurement is complete. If you are using the USB cable and PC software to collect data or the optional printer, you will see the results of individual measurement rounds. The display of the Sigma 702ET instrument will show the average of the measurement rounds and the sample standard deviation of the results.

As the ring is pulled through a surface, a curve is plotted, showing the tension value calculated from the balance reading at each instant. For a single measurement, the highest value obtained will be the result of the measurement. For multi-round measurements, the results are averaged, and their sample standard deviation calculated and displayed.

10. CLEANING PROCEDURE

In all measurements done with Sigma 702ET, the cleanliness of the used probes and beakers is extremely important for reliable and reproducible results. The best cleaning procedure depends on the samples used. Here is a typical cleaning procedure when water or oil is used.

- **Cleaning of the glass beaker:**
Rinse three times with n-heptane (only after preceding tests with oil), afterwards with 2-propanol (also in case of unused beakers and after water testing). Rinse with hot tap water and afterwards thoroughly with deionized water/bi-distilled water.
- **Cleaning of the platinum probe:**
Rinse three times with n-heptane; afterwards with deionized water and heat it in the oxidizing flame for approx. 5 s in an ethanol or natural gas burner to red heat.

REFERENCE

[1] Manufacturer Instruction

Doc.: EOP [6]-04-E	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 5 of 5	